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LOS Assignment Theory-4

Hard link vs. soft link

Part A — Create the links

Create a directory named link_lab and navigate to it.

Create a file named security.txt containing the following text:

Linux File System Security Lab

Confidential Security Report

Create:

a hard link named security_hard.txt

a symbolic link named security_soft.txt

List all three files using a command that displays their inode numbers.

Record the inode numbers of:

security.txt

security_hard.txt

security_soft.txt

What difference do you observe?

```
Konsole File Edit View Bookmarks Plugins Settings Help Search (Friday, Sep 11)
anonymous@anonymous-supreme in ~ via v3.14.7
# ) mkdir link_lab

anonymous@anonymous-supreme in ~ via v3.14.7
# ) cd link_lab

anonymous@anonymous-supreme in ~/link_lab
# ) echo -e "Linux File System Security Lab\nConfidential Security Report" > security.txt

anonymous@anonymous-supreme in ~/link_lab
# ) cat security.txt
Linux File System Security Lab
Confidential Security Report

anonymous@anonymous-supreme in ~/link_lab
# ) ln security.txt security_hard.txt

anonymous@anonymous-supreme in ~/link_lab
# ) ln -s security.txt security_soft.txt

anonymous@anonymous-supreme in ~/link_lab
# ) ls -li
5562070 .rw-r--r-- 60 anonymous 11 Sep 17:50 security.txt
5562070 .rw-r--r-- 60 anonymous 11 Sep 17:50 security_hard.txt
5562092 lrwxrwxrwx - anonymous 11 Sep 17:50 security_soft.txt -> security.txt

anonymous@anonymous-supreme in ~/link_lab
# ) _
```

Observation

security.txt and security_hard.txt have the same inode number, while security_soft.txt has a different inode number. This shows that the hard link points to the same file, whereas the soft link is a separate file.

Part B — Examine File Metadata

Use `ls -l` to examine the three files. Identify the file type of each file from the first character of the permission field.

Compare the link count of the original file, hard link, and symbolic link. Explain the observed values.

Use the `stat` command on all three files and record the following information:

Parameter	Original Hard Link	Soft Link
-----------	--------------------	-----------

Inode number

File type

Size

Link count

Permissions

Compare the file sizes of the original file, hard link, and symbolic link. Why might the size of the symbolic link differ from the original file?

```
anonymous@anonymous-supreme in ~/link_lab
/ ) ls -l security.txt security_hard.txt security_soft.txt
-rw-r--r-- 60 anonymous 11 Sep 17:50 security.txt
-rw-r--r-- 60 anonymous 11 Sep 17:50 security_hard.txt
lrwxrwxrwx - anonymous 11 Sep 17:50 security_soft.txt -> security.txt

anonymous@anonymous-supreme in ~/link_lab
/ ) stat security.txt
File: security.txt
Size: 60          Blocks: 8          IO Block: 4096   regular file
Device: 0,51     Inode: 5562070    Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/anonymous)   Gid: ( 1000/anonymous)
Access: 2026-09-11 17:50:17.675792713 +0530
Modify: 2026-09-11 17:50:17.675792713 +0530
Change: 2026-09-11 17:50:37.574132951 +0530
Birth: 2026-09-11 17:50:17.675792713 +0530

anonymous@anonymous-supreme in ~/link_lab
/ ) stat security_hard.txt
File: security_hard.txt
Size: 60          Blocks: 8          IO Block: 4096   regular file
Device: 0,51     Inode: 5562070    Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/anonymous)   Gid: ( 1000/anonymous)
Access: 2026-09-11 17:50:17.675792713 +0530
Modify: 2026-09-11 17:50:17.675792713 +0530
Change: 2026-09-11 17:50:37.574132951 +0530
Birth: 2026-09-11 17:50:17.675792713 +0530

anonymous@anonymous-supreme in ~/link_lab
/ ) stat security_soft.txt
File: security_soft.txt -> security.txt
Size: 12          Blocks: 8          IO Block: 4096   symbolic link
Device: 0,51     Inode: 5562092    Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/anonymous)   Gid: ( 1000/anonymous)
Access: 2026-09-11 17:50:46.075514588 +0530
Modify: 2026-09-11 17:50:46.075514588 +0530
Change: 2026-09-11 17:50:46.075514588 +0530
Birth: 2026-09-11 17:50:46.075514588 +0530

anonymous@anonymous-supreme in ~/link_lab
/ )
```

3. File size observation

```
anonymous@anonymous-supreme in ~/link_lab
# ) readlink security_soft.txt
security.txt

anonymous@anonymous-supreme in ~/link_lab
# ) readlink security_hard.txt

anonymous@anonymous-supreme in ~/link_lab
[*] # readlink -f security_soft.txt
/home/anonymous/link_lab/security.txt

anonymous@anonymous-supreme in ~/link_lab
# )
```

The original file and hard link have the same size because they point to the same file. The soft link has a different and smaller size because it only stores the path of the original file.

Part C — Investigate readlink

Execute readlink on security_soft.txt. What information is displayed?

Execute readlink on security_hard.txt. What do you observe?

Based on your observations, explain why readlink works differently for hard links and symbolic links.

- A symbolic link stores the path of another file, while a hard link directly points to the same inode.

Use:

```
readlink -f security_soft.txt
```

What does the -f option reveal?

- readlink displays the target path of a symbolic link, but gives no output for a hard link. The -f option shows the absolute path of the target file.

Part D — Content Modification

Append the following line to security.txt:

Unauthorized login detected

Display the contents using all three filenames.

Is the newly added content visible through both links? Explain why.

Modify the file using security_hard.txt. Check the contents through security.txt and security_soft.txt. What do you observe?

```
anonymous@anonymous-supreme in ~/link_lab
/ ) echo "Unauthorized login detected" >> security.txt

anonymous@anonymous-supreme in ~/link_lab
/ ) cat security.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected

anonymous@anonymous-supreme in ~/link_lab
/ ) cat security_hard.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected

anonymous@anonymous-supreme in ~/link_lab
/ ) cat security_soft.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected

anonymous@anonymous-supreme in ~/link_lab
/ ) _
```

```
anonymous@anonymous-supreme in ~/link_lab
# ) echo "Investigation started" >> security_hard.txt

anonymous@anonymous-supreme in ~/link_lab
# ) cat security.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected
Investigation started
Investigation started
Investigation started

anonymous@anonymous-supreme in ~/link_lab
# ) cat security_soft.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected
Investigation started
Investigation started
Investigation started

anonymous@anonymous-supreme in ~/link_lab
# ) _
```

Observation

The new content is also visible through security.txt and security_soft.txt. This is because security_hard.txt refers to the same file as security.txt, and the soft link points to that file.

Part E — Rename Experiment

Rename:

security.txt

to:

incident.txt

Try accessing the content through security_hard.txt. Does it still work?

Try accessing the content through security_soft.txt. Does it still work?

Use readlink security_soft.txt after renaming the original file. What path is still stored in the symbolic link?

Explain why renaming the original file affects the symbolic link differently from the hard link.

```
anonymous@anonymous-supreme in ~/link_lab
# ) mv security.txt incident.txt

anonymous@anonymous-supreme in ~/link_lab
# ) cat security_hard.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected
Investigation started
Investigation started
Investigation started

anonymous@anonymous-supreme in ~/link_lab
# ) cat security_soft.txt
cat: security_soft.txt: No such file or directory

anonymous@anonymous-supreme in ~/link_lab
[*] # readlink security_soft.txt
security.txt

anonymous@anonymous-supreme in ~/link_lab
# ) _
```

Observation

The hard link still works after renaming because it points to the same inode. The soft link does not work because it stores the filename security.txt, which no longer exists after renaming it to incident.txt.

Part F — Deletion Experiment

Recreate the links if necessary and then perform the following.

Delete the original file.

Check whether security_hard.txt can still be accessed.

Check whether security_soft.txt can still be accessed.

Use ls -l and readlink to investigate the symbolic link after deleting its target.

```
anonymous@anonymous-supreme in ~/link_lab
[?] rm security.txt
rm: cannot remove 'security.txt': No such file or directory

anonymous@anonymous-supreme in ~/link_lab
[*] [?] cat security_hard.txt
Linux File System Security Lab
Confidential Security Report
Unauthorized login detected
Investigation started
Investigation started
Investigation started

anonymous@anonymous-supreme in ~/link_lab
[?] cat security_soft.txt
cat: security_soft.txt: No such file or directory

anonymous@anonymous-supreme in ~/link_lab
[*] [?] ls -l security_soft.txt
lrwxrwxrwx - anonymous 11 Sep 17:50 security_soft.txt -> security.txt

anonymous@anonymous-supreme in ~/link_lab
[?] readlink security_soft.txt
security.txt

anonymous@anonymous-supreme in ~/link_lab
[?] ls
-rw-r--r-- 154 anonymous 11 Sep 18:38 incident.txt
-rw-r--r-- 154 anonymous 11 Sep 18:38 security_hard.txt
lrwxrwxrwx - anonymous 11 Sep 17:50 security_soft.txt -> security.txt

anonymous@anonymous-supreme in ~/link_lab
[?] )
```

What is a dangling/broken symbolic link?

- A dangling symbolic link is a link whose target file has been deleted or is no longer available.

Does deleting the original filename immediately delete the underlying data when a hard link exists? Justify your answer using the link count and inode.

- No. The data is still available through security_hard.txt because the hard link points to the same inode. After deleting security.txt, the link count decreases from **2 to 1**, so the data remains. The data is removed only when the link count becomes **0** and no process is using the file.

Create a directory named evidence. Attempt to create a hard link to this directory. Record the result.

Create a symbolic link named evidence_link pointing to the evidence directory. Was the operation successful?

Based on your experiment, compare hard and symbolic links with respect to directory linking.

If multiple filesystems/mount points are available in the laboratory environment, attempt to create:

a hard link across filesystems

a symbolic link across filesystems

Final Comparison

Based only on your experimental observations, complete the following table:

Parameter	Hard Link	Symbolic Link
-----------	-----------	---------------

Inode number		
--------------	--	--

File type		
-----------	--	--

Link count		
------------	--	--

File size		
-----------	--	--

Stores target pathname?		
-------------------------	--	--

readlink output		
-----------------	--	--

Effect of modifying target		
----------------------------	--	--

Effect of renaming target		
---------------------------	--	--

Effect of deleting target		
---------------------------	--	--

Can become a dangling or broken link?		
---------------------------------------	--	--

Directory linking		
-------------------	--	--

Cross-filesystem linking		
--------------------------	--	--

sort		
------	--	--

```
anonymous@anonymous-supreme in ~/link_lab
# ) mkdir evidence

anonymous@anonymous-supreme in ~/link_lab
# ) ln evidence evidence_hard
ln: evidence: hard link not allowed for directory

anonymous@anonymous-supreme in ~/link_lab
[*] # ln -s evidence evidence_link

anonymous@anonymous-supreme in ~/link_lab
# ) ls -l
drwxr-xr-x  - anonymous 11 Sep 19:14 evidence
lrwxrwxrwx  - anonymous 11 Sep 19:15 evidence_link -> evidence
-rw-r--r-- 154 anonymous 11 Sep 18:38 incident.txt
-rw-r--r-- 154 anonymous 11 Sep 18:38 security_hard.txt
lrwxrwxrwx  - anonymous 11 Sep 17:50 security_soft.txt -> security.txt

anonymous@anonymous-supreme in ~/link_lab
```

Observation

Hard links cannot normally be created for directories, while symbolic links can point to directories.

```

anonymous@anonymous-supreme in ~/link_lab
# ) ls -l
drwxr-xr-x - anonymous 11 Sep 19:14 evidence
lrwxrwxrwx - anonymous 11 Sep 19:15 evidence_link -> evidence
.rw-r--r-- 154 anonymous 11 Sep 18:38 incident.txt
.rw-r--r-- 154 anonymous 11 Sep 18:38 security_hard.txt
lrwxrwxrwx - anonymous 11 Sep 17:50 security_soft.txt -> security.txt

anonymous@anonymous-supreme in ~/link_lab
# ) df -T
Filesystem      Type      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /
devtmpfs        devtmpfs   3877792      0      3877792  0% /dev
tmpfs           tmpfs      3991448      23512   3967936  1% /dev/shm
efivarfs        efivarfs   184          104      76     58% /sys/firmware/efi/efivars
tmpfs           tmpfs      1596580      1912   1594668  1% /run
none            tmpfs      1024         0       1024    0% /run/credentials/systemd-journald.service
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /home
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /root
tmpfs           tmpfs      3991448      5580   3985868  1% /tmp
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /var/log
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /var/cache
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /srv
/dev/nvme0n1p5 btrfs     326281700 140938720 180929024 44% /var/tmp
/dev/nvme0n1p1 vfat      262144      75908   186236  29% /boot/efi
tmpfs           tmpfs      798288       96     798192  1% /run/user/1000

anonymous@anonymous-supreme in ~/link_lab
# ) ln test.txt /path/on/another/filesystem/test_hard.txt
ln: failed to access 'test.txt': No such file or directory

anonymous@anonymous-supreme in ~/link_lab
[*] # ln -s /path/on/another/filesystem/test.txt test_soft.txt

anonymous@anonymous-supreme in ~/link_lab
# ) _

```

Observation

A hard link cannot normally cross filesystems because it points directly to an inode. A symbolic link can cross filesystems because it stores the pathname of the target.

Parameter	Hard Link	Symbolic Link
Inode number	Same as original	Different
File type	Regular file	Symbolic link

Link count	Increases original link count	Usually 1
File size	Same as original	Size of stored pathname
Stores target pathname?	No	Yes
readlink output	No output	Shows target pathname
Effect of modifying target	Changes the same data	Changes are visible through link
Effect of renaming target	Still works	Usually stops working
Effect of deleting target	Still works if another hard link exists	Becomes broken
Can become dangling/broken?	No	Yes
Directory linking	Not normally allowed	Allowed
Cross-filesystem linking	Not allowed	Allowed

Suppose alerts.txt contains:

192.168.1.10 admin 5

10.0.0.8 root 12

172.16.0.5 guest 2

10.0.0.20 admin 8

Format:

IP-address username failed attempts

Sort according to failed attempts

Sort a list of suspicious IP addresses alphabetically.

Display only unique suspicious IP addresses using sort.

Sort a list of detected port numbers numerically.

Display port numbers in descending numerical order.

Sort usernames without considering uppercase/lowercase differences.

From a failed-login log, count how many times each IP address occurs and display the most frequent IP first.

Given IP,username,failed_attempts, sort records based on the number of failed attempts.

Sort a CSV vulnerability report based on severity score.

Extract IP addresses using grep, then use sort and uniq to identify unique IP addresses.

Create a pipeline that displays the top 5 IP addresses responsible for the most failed login attempts.

```
anonymous@anonymous-supreme in ~/link_lab (Friday, Sept 11 7:4)
/ ) nano alerts.txt

anonymous@anonymous-supreme in ~/link_lab took 21s565ms
/ ) cat alerts.txt
GNU nano 9.2 alerts.txt
192.168.1.10 admin 5
10.0.0.8 root 12
172.16.0.5 guest 2
10.0.0.20 admin 8192.168.1.10 admin 5
10.0.0.8 root 12
172.16.0.5 guest 2
10.0.0.20 admin 8

anonymous@anonymous-supreme in ~/link_lab
/ ) sort -k3,3n alerts.txt
172.16.0.5 guest 2
172.16.0.5 guest 2
192.168.1.10 admin 5
10.0.0.20 admin 8
GNU nano 9.2 alerts.txt
10.0.0.8 root 12
10.0.0.8 root 12
10.0.0.20 admin 8192.168.1.10 admin 5

anonymous@anonymous-supreme in ~/link_lab
/ ) nano suspicious.txt

anonymous@anonymous-supreme in ~/link_lab took 9s578ms
/ ) sort suspicious.txt
10.0.0.20 admin 8
10.0.0.20 admin 8
10.0.0.8 root 12
10.0.0.8 root 12
172.16.0.5 guest 2
172.16.0.5 guest 2
192.168.1.10 admin 5
192.168.1.10 admin 5
GNU nano 9.2 suspicious.txt

anonymous@anonymous-supreme in ~/link_lab
/ ) sort -u suspicious.txt
10.0.0.20 admin 8
10.0.0.8 root 12
172.16.0.5 guest 2
192.168.1.10 admin 5
GNU nano 9.2 suspicious.txt
```

```
anonymous@anonymous-supreme in ~/link_lab took 8s33ms
# ) sort -n ports.txt
GNU nano 9.2 ports.txt
21
21
22
22
80
80
443
443
8080
8080

anonymous@anonymous-supreme in ~/link_lab
# ) sort -nr ports.txt
8080
8080
443
443
80
80
22
22
21
21
GNU nano 9.2 ports.txt

anonymous@anonymous-supreme in ~/link_lab
# ) nano usernames.txt

anonymous@anonymous-supreme in ~/link_lab took 6s353ms
# ) sort -f usernames.txt
admin
ADMIN
Ayush
guest
Guest
root
Root

anonymous@anonymous-supreme in ~/link_lab
# ) nano failed.log

anonymous@anonymous-supreme in ~/link_lab took 5s232ms
# ) grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' failed.log | sort -u
10.0.0.20
```

```

anonymous@anonymous-supreme in ~/link_lab took 5s232ms
/ ) grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' failed.log | sort -u
10.0.0.20
10.0.0.8
172.16.0.5
192.168.1.10

anonymous@anonymous-supreme in ~/link_lab
/ ) grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' failed.log | sort | uniq -c | sort -nr
5 192.168.1.10
4 10.0.0.8
2 172.16.0.5
1 10.0.0.20

anonymous@anonymous-supreme in ~/link_lab
/ ) grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' failed.log | sort | uniq -c | sort -nr | head -5
5 192.168.1.10
4 10.0.0.8
2 172.16.0.5
1 10.0.0.20

anonymous@anonymous-supreme in ~/link_lab
/ ) nano login.csv

anonymous@anonymous-supreme in ~/link_lab took 7s600ms
/ ) sort -t',' -k3,3n login.csv
172.16.0.5,guest,2
192.168.1.10,admin,5
10.0.0.20,admin,8
10.0.0.8,root,12
192.168.1.10,admin,15

anonymous@anonymous-supreme in ~/link_lab
/ ) sort -t',' -k3,3nr login.csv
192.168.1.10,admin,15
10.0.0.8,root,12
10.0.0.20,admin,8
192.168.1.10,admin,5
172.16.0.5,guest,2

anonymous@anonymous-supreme in ~/link_lab
/ ) nano vulnerabilities.csv

anonymous@anonymous-supreme in ~/link_lab took 5s356ms
/ ) sort -t',' -k3,3n vulnerabilities.csv
CVE-2026-003,PHP,5.6
CVE-2026-005,nginx,6.4
CVE-2026-001,Apache,7.5
CVE-2026-004,MySQL,8.1

```

```
anonymous@anonymous-supreme in ~/link_lab took 5s356ms  
✎ ) sort -t',' -k3,3n vulnerabilities.csv
```

```
CVE-2026-003,PHP,5.6  
CVE-2026-005,nginx,6.4  
CVE-2026-001,Apache,7.5  
CVE-2026-004,MySQL,8.1  
CVE-2026-002,OpenSSH,9.8
```

```
anonymous@anonymous-supreme in ~/link_lab  
✎ ) sort -t',' -k3,3nr vulnerabilities.csv
```

```
CVE-2026-002,OpenSSH,9.8  
CVE-2026-004,MySQL,8.1  
CVE-2026-001,Apache,7.5  
CVE-2026-005,nginx,6.4  
CVE-2026-003,PHP,5.6
```